

Visual Inspection Microscopes

Visual inspection is a common method for fault detection and quality control in the electronics industry. As electronic assemblies continue to shrink in size, it becomes impossible to reach out various inspection points defined by standards such as IPC-A-610E with direct human vision. Magnification aids such as inspection microscopes can increase the efficiency and accuracy of visual inspection to a great extent



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No manufacturing process is perfect and there is always a percentage of boards with defects. These boards need to be repaired or discarded to ensure the quality of the end product. Various inspection processes are designed to efficiently find out these defective boards and segregate them from the good ones. Most defects, such as misaligned components, wrong orientation, solder bridging, open connection, improper soldering and physical damage can be easily detected by visual inspection. There are certain defect types that can only be discovered by more sophisticated techniques such as X-ray imaging. But visual inspection remains

the primary means of finding defects in circuit assemblies and correcting them before they reach later stages of test.

Visual inspection standard

IPC-A-610E standard, which is a collection of visual quality acceptability requirements for electronic assemblies, also indicates the need for magnification aids and acceptable parameters. Reproduced below is the 'Magnification Aids' section from IPC-A-610E standard:

1.9 Magnification Aids: For visual inspection, some individual specifications may call for magnification aids for examining printed board assemblies.

The tolerance for magnification aids is $\pm 15\%$ of the selected magnification power. Magnification aids, if used for inspection, shall be appropriate with the item being

inspected. Unless magnification requirements are otherwise specified by contractual documentation, the magnifications in Table I and Table II are determined by the item being inspected.

Referee conditions are used to verify product rejected at the inspection magnification power. For assemblies with mixed land widths, the greater magnification may be used for the entire assembly.

Illumination

The human eye only sees the information carried by reflected light. Therefore illumination of the object is as important as magnifying it. The IPC-A-610E standard does not give much detail about the illumination. Reproduced below is the 'Lighting' section from the standard:

1.10 Lighting: Lighting shall be adequate for the item being inspected.

Illumination at the surface of workstations should be at least 1000 lm/m^2 (approximately 93-foot candles). Light sources should be selected to prevent shadows.

Note: In selecting a light source, the colour temperature of the light is an important consideration. Light ranges from 3000 to 5000°K enable users to distinguish various printed circuit assembly features and contaminates with increased clarity.

There are several types of microscope illumination. The basic type mounts on the microscope itself and has a hood over a circular fluorescent bulb. The circular shape distributes illumination evenly. Fluorescent lamps have long been used for such applications but the greenish spectrum of light emitted by them, besides flickering, shadows and humming caused by the